

Maths Class: _____

20 MATHS

2005

Instructions:

- Write your name and class at the top of this page, and at the top of each answer sheet.
- At the end of the examination this examination paper must be attached to the front of your answers.
- All questions are of equal value and may be attempted.
- All necessary working must be shown. Marks will be deducted for careless or badly arranged work.
- Marks indicated are a guide only and may be varied if necessary
- Non programmable calculators may be used.

(For markers use only)

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Total
										120

Question 1 (12 marks)

- a) Evaluate $\sin^2 1.7 + \cos^2 1.7$ (1)
- b) Factorise fully $x^4 - 16$ (2)
- c) Find rational numbers a and b such that $4a + b\sqrt{7} = 36 + \sqrt{63}$ (2)
- d) For $3^x = 7$, find x correct to two decimal places (2)
- e) Solve $\frac{5}{x} = \frac{4}{x-3}$ (2)
- f) i) Sketch $y = x^2 - x - 2$
- ii) Hence solve $x^2 - x - 2 < 0$ (3)

Question 2 (12 marks) (Start a new page)

A, B and C are the points (0,4), (1,-5) and (6,-2) respectively.

- a) Sketch the triangle ABC on a number plane (1)
- b) Show that the length of AC is $6\sqrt{2}$ units. (1)
- c) Show that the line AC has equation $x + y - 4 = 0$ (2)
- d) Show that the perpendicular distance from B to AC is $4\sqrt{2}$ units (2)
- e) Hence find the area of triangle ABC (1)
- f) Find the co-ordinates of D if C is the midpoint of the interval BD (2)
- g) Find the length of the interval AB (1)
- h) Hence or otherwise find to the nearest degree the size of $\angle BAC$ (2)

Question 3 (12 marks) Start a new page

- a) Consider the function $y = x^3 + x$
- i) Find $\frac{dy}{dx}$ (1)
- ii) Hence explain why the function is increasing for all values of x (1)

b) Differentiate with respect to x

i) $\frac{3}{x^2}$ (1)

ii) $\frac{\cos 3x}{x}$ (2)

iii) $\frac{3}{x^2 + 1}$ (2)

c) Find $\int \frac{1}{\sqrt{2x+1}} dx$ (3)

d) Solve $\log_5 3 = 2 \log_5 6 - \log_5 x$ (2)

Question 4 (12 marks) Start a new page

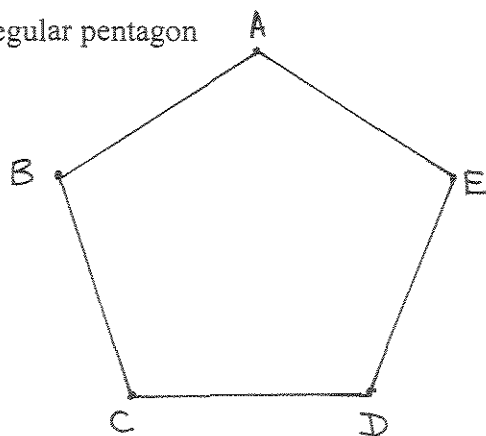
a) An aircraft flies 300 nautical miles from its base B on a bearing of 050° to point Q. It then flies 225 nautical miles due north to point P.

i) Draw a diagram to show this information and find angle $\angle BQP$. (2)

ii) Calculate the distance from the base B to point P to the nearest nautical mile. (2)

iii) Find the direction the plane must now fly to return to its base (2)

b) ABCDE is a regular pentagon



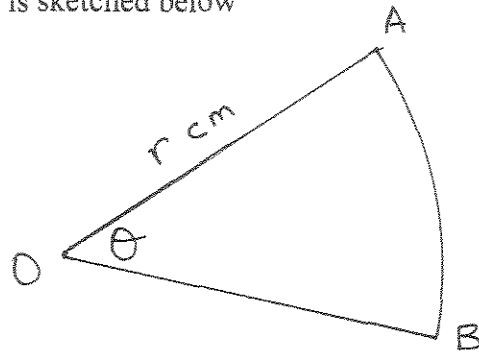
i) Find the size of $\angle ABC$ (1)

ii) Prove $\triangle ABC \equiv \triangle AED$ (3)

iii) Find $\angle CAD$ giving reasons for your answer (2)

Question 5 (12 marks) (Start a new page)

a) The sector OAB is sketched below



- i) Write a formula for the arc length AB in terms of r and θ (1)
- ii) Write a formula for the area of the sector OAB in terms of r and θ (1)
- iii) If the sector has area $\pi \text{ cm}^2$ and the arc length AB is $\frac{\pi}{4} \text{ cm}$ find r and θ (2)

b) Solve $2 \sin 2\theta - 1 = 0$ for $0^\circ \leq \theta \leq 360^\circ$ (4)

c) i) Sketch $y = \log_{10} x$ (1)

ii) Complete the table below for $y = \log_{10} x$
(leave answers to 3 decimal places) (1)

x	1	1.5	2	2.5	3
y					

iii) Use Simpsons Rule with five function values from part ii) to estimate

$$\int_1^3 \log_{10} x \, dx$$

Give your answer to 2 decimal places (2)

Question 6 (12 marks) (Start a new page)

a) Let α and β be the roots of the equation $2x^2 - 3x - 4 = 0$. find the values of

i) $\alpha + \beta$ (1)

ii) $\alpha\beta$ (1)

iii) $(\alpha + 1)(\beta + 1)$ (1)

iv) $\alpha^2 + \beta^2$ (1)

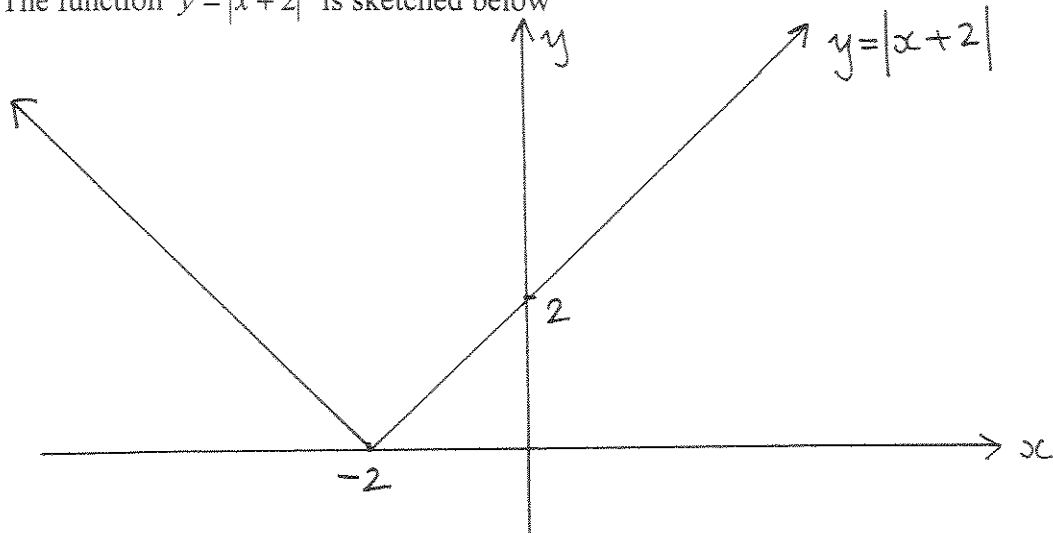
b) A parabola has equation $y^2 - 6y - 3 = 12x$

i) Write the parabola in the form $(y - k)^2 = 4a(x - h)$ (1)

ii) Find the co-ordinates of the vertex. (1)

iii) Sketch the parabola showing the co-ordinates of the focus. (2)

c) The function $y = |x + 2|$ is sketched below



i) Copy the graph onto your answer paper and sketch $y = 2x + 6$ on the same number plane (1)

ii) Show the co-ordinates of the point of intersection are $(-2\frac{2}{3}, \frac{2}{3})$ (2)

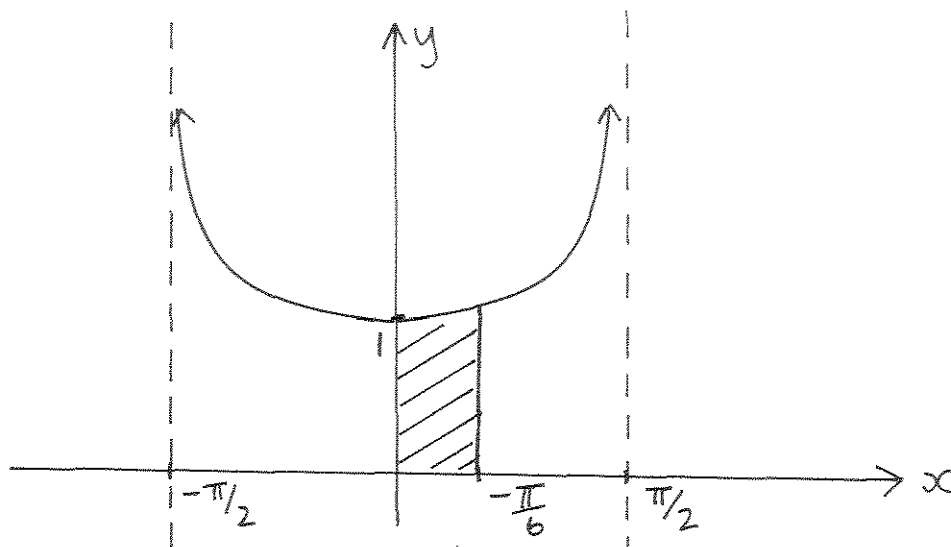
iii) Hence or otherwise solve $|x + 2| > 2x + 6$ (1)

Question 7 (12 marks) (Start a new page)

- a) i) Sketch $y=3\cos 2x$ for $0 \leq x \leq 2\pi$ (2)
- ii) From the sketch, state how many points of inflexion the curve has in the domain $0 \leq x \leq 2\pi$ (1)

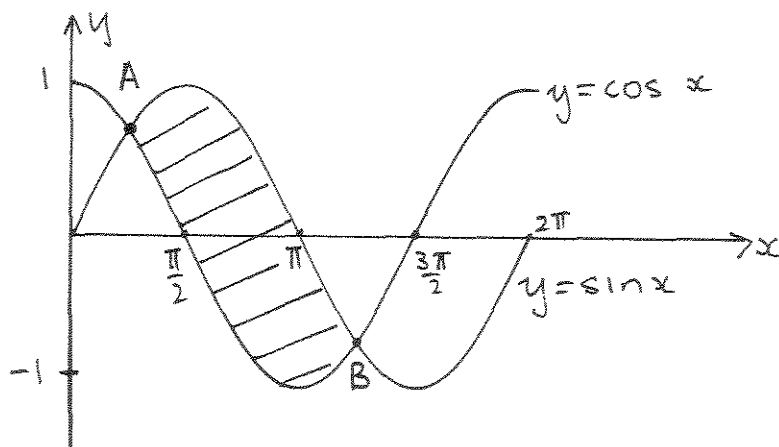
- b) The diagram below shows the region between the curve $y = \sec x$ and the x axis from $x = 0$ to $x = \frac{\pi}{6}$

Find the volume generated when this region is rotated around the x axis



(3)

- c) The curves $y = \sin x$ and $y = \cos x$ are sketched below for $0 \leq x \leq 2\pi$



- i) Show the x co-ordinates of the points of intersection
A and B are $\frac{\pi}{4}$ and $\frac{5\pi}{4}$ respectively. (2)
- ii) Hence find the shaded area. (4)

Question 8 (12 marks) (Start a new page)

- a) The function $y = x^3 - 3x^2 - 9x + 1$ is defined in the domain $-4 \leq x \leq 5$.
- i) Find the co-ordinates of any turning points and determine their nature. (3)
 - ii) Find the co-ordinates of any points of inflexion. (1)
 - iii) Sketch the function in the domain and label stationary points, points of inflexion and end points. (3)
 - iv) Determine the minimum value of the function y in the given domain. (1)
- b) For the function $y = kx^2 - 4\sqrt{3}x + k - 1$
- i) Find an expression for the discriminant. (1)
 - ii) For what values of k is the function positive definite? (3)

Question 9 (12 marks) (Start a new page)

- a) i) Find $\frac{d}{dx}(\sin^3 x)$ (2)
- ii) Hence find $\int \cos x \cdot \sin^2 x \, dx$ (1)
- b) Alan borrows \$130,000 to start a signwriting business. He is charged interest on the balance owing at the rate of 9.75%pa compounded monthly and agrees to repay the loan including interest by making equal monthly instalments of \$M.
- i) How much does Alan owe at the end of the first month just before he makes an instalment? (1)
 - ii) Write an expression involving M for the total amount owed by Alan just after the second instalment is paid. (2)
 - iii) Calculate the value of M (to the nearest cent) which will repay the loan after 13 yrs. (4)
 - iv) In how many months (to the nearest whole month) will the loan be repaid if Alan made instalments of \$1700 per month (2)

Question 10 (12 marks) (start a new page)

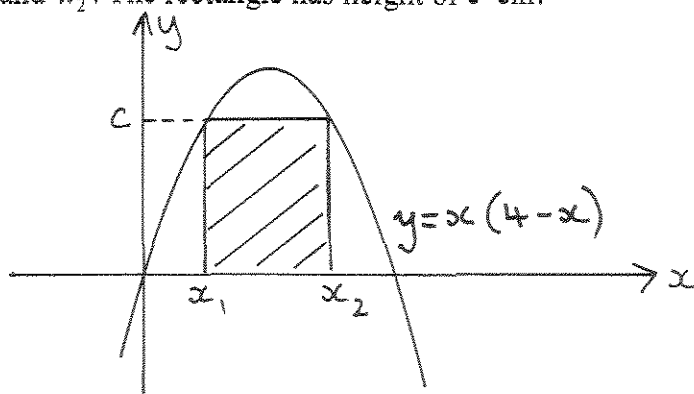
- a) In a new Quiz show of “The Sky is the Limit”, you win \$6 000 for answering the first question correctly, \$14 000 for answering the second question correctly, \$22 000 for answering the third question correctly and so on for the following questions. The prizes form an arithmetic sequence.

You finish when you answer a question incorrectly. Your total winnings for the contest is the sum of money you win on each question.

- i) What is the prize money for the 10th question only? (2)
- ii) How many questions must you correctly answer to exceed \$1 000 000 in total winnings? (4)

- b) A rectangle has two vertices on the curve $y = x(4 - x)$.

The other two vertices are on the x axis in the interval $0 \leq x \leq 4$ and are called x_1 and x_2 . The rectangle has height of c cm.



- i) Find the length $(x_2 - x_1)$ in terms of c and hence show the area of the rectangle is given by $A = 2c\sqrt{4 - c}$ cm² (3)
- ii) Show the maximum area of this rectangle is $\frac{32\sqrt{3}}{9}$ cm². (3)

(showing marks)

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QUESTION 1

a) $\sin^2 1.7 + \cos^2 1.7 = \underline{1}$ — (1)

b) $x^4 - 16 = (x^2 - 4)(x^2 + 4)$ — (2)
 $= (x - 2)(x + 2)(x^2 + 4)$

c) $4a + b\sqrt{7} = 36 + \sqrt{63}$
 $= 36 + 3\sqrt{7}$

$\therefore 4a = 36$

$a = 9$

$b = 3$ — (2)

d) $3^x = 7$

$\log_{10} 3^x = \log_{10} 7$

$x \log_{10} 3 = \log_{10} 7$

$x = \frac{\log_{10} 7}{\log_{10} 3}$

$x = 1.77$ — (2)

e) $\frac{5}{x} = \frac{4}{x-3}$

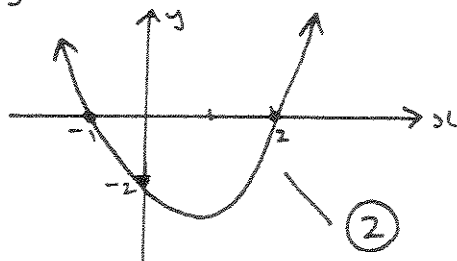
$4x = 5(x-3)$

$4x = 5x - 15$

$\therefore x = 15$ — (2)

f) i) $y = x^2 - x - 2$

$y = (x-2)(x+1)$



ii)

$\therefore x^2 - x - 2 < 0$

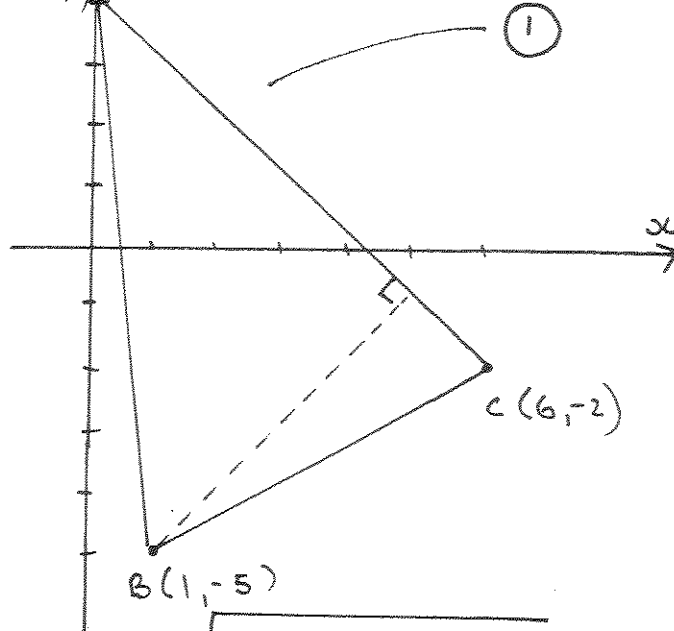
for $\underline{-2 < x < 1}$

(1)

QUESTION 2

a)

A(0,4)



b)

$AC = \sqrt{(0-6)^2 + (4+2)^2}$
 $= \sqrt{36+36}$
 $= \sqrt{72}$
 $= 6\sqrt{2}$ units — (1)

c)

AC has gradient $m = \frac{4-(-2)}{0-6} = -1$
and y intercept $b = 4$

$\therefore$ eqn AC: $y = -x + 4$

$\therefore x + y - 4 = 0$ — (2)

d)

$p = \left| \frac{1 \times 1 + 1 \times (-5) - 4}{\sqrt{1+1}} \right|$

$p = \left| \frac{-8}{\sqrt{2}} \right|$

$p = \frac{8}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$

$p = 4\sqrt{2}$ units — (2)

e) Area $\triangle ABC = \frac{6\sqrt{2} \times 4\sqrt{2}}{2}$

$= 24 \text{ unit}^2$ — (1)

f) Let $D(x, y)$

$$\therefore \frac{x+1}{2} = 6$$

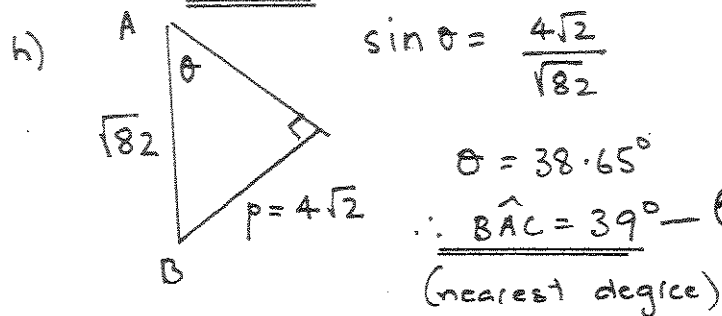
$$x+1=12$$

$$x=11$$

$$\therefore D(11, 1) \text{ ————— } \textcircled{2}$$

g) $AB = \sqrt{(0-1)^2 + (4+5)^2}$

$$= \sqrt{82} \text{ units ————— } \textcircled{1}$$



QUESTION 3

a) i) $\frac{dy}{dx} = 3x^2 + 1 \text{ ————— } \textcircled{1}$

ii) for all values of x , $x^2 \geq 0$

$\therefore \frac{dy}{dx} > 0 \therefore$ increasing for all x ————— $\textcircled{1}$

b) i) $\frac{d}{dx} (3x^{-2}) = -6x^{-3}$

$$= \frac{-6}{x^3} \text{ ————— } \textcircled{1}$$

ii) $u = \cos 3x \quad v = x$

$$u' = -3 \sin 3x \quad v' = 1 \text{ ————— } \textcircled{2}$$

$$\frac{dy}{dx} = \frac{-3x \sin 3x - \cos 3x}{x^2}$$

iii) $\frac{d}{dx} (3(x^2+1)^{-1})$

$$= -3 \times 2x (x^2+1)^{-2}$$

$$= \frac{-6x}{(x^2+1)^2} \text{ ————— } \textcircled{2}$$

c) $\int (2x+1)^{-1/2} dx = \frac{(2x+1)^{1/2}}{2 \times \frac{1}{2}} + c$

take off 'c' for no

$$= \sqrt{2x+1} + c \text{ ————— } \textcircled{3}$$

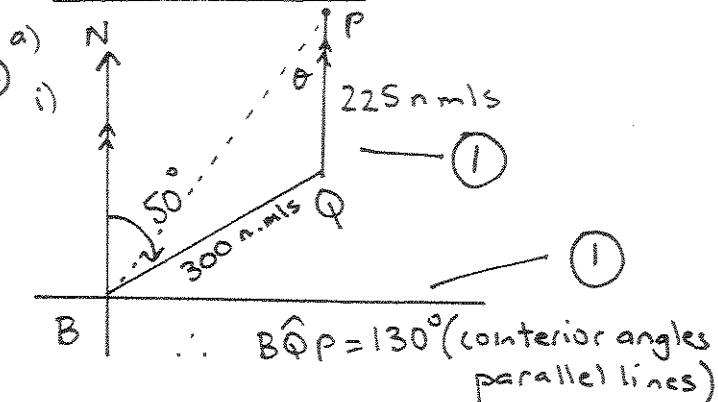
d) $\log_5 3 = 2 \log_5 6 - \log_5 x$

$$\log_5 3 = \log_5 \left(\frac{36}{x} \right)$$

$$3 = \frac{36}{x}$$

$$x = 12 \text{ ————— } \textcircled{2}$$

QUESTION 4



ii) $BP^2 = 300^2 + 225^2 - 2 \times 300 \times 225 \cos 130^\circ$

$$BP = 477 \text{ nat mls (to nearest n.ml) ————— } \textcircled{2}$$

iii) Let $\hat{BPQ} = \theta$

$$\frac{\sin \theta}{300} = \frac{\sin 130^\circ}{477}$$

$$\sin \theta = \frac{300 \sin 130^\circ}{477}$$

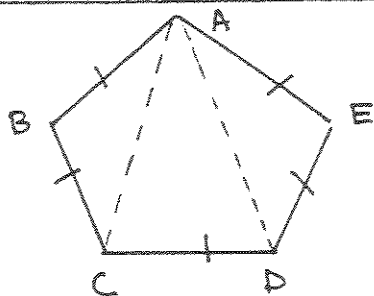
$$\theta = 28.8^\circ \text{ ————— } \textcircled{1}$$

$\therefore$ Bearing $180^\circ + 28.8^\circ \text{ ————— } \textcircled{1}$

$$= 209^\circ \text{ (to nearest degree)}$$

OR $\underline{\underline{S 29^\circ W}}$

b)



①

i) Angle Sum Pentagon = 3×180
 $\therefore \hat{ABC} = \frac{540^\circ}{5} = 108^\circ$ (angle of regular pentagon)

ii) In $\triangle ABC$ and $\triangle AED$
 $\hat{ABC} = \hat{AED}$ (angles in regular pentagon)
 $AB = AE$ (sides of regular pentagon)
 $BC = DE$ (sides of regular pentagon)
 $\therefore \triangle ABC \equiv \triangle AED$ (SAS)

iii) $\hat{BAC} = 36^\circ$ (angle sum isosceles triangle)

$\hat{DAE} = 36^\circ$ (corresp. angles in congruent triangles)

since $\hat{BAE} = 108^\circ$ (angle of regular pentagon)

$\therefore \hat{CAD} = 108^\circ - 2 \times 36^\circ$

② $= 36^\circ$

QUESTION 5

a) i) $AB = r\theta$ ——— ①
 ii) $\text{Area Sector} = \frac{1}{2} r^2 \theta$ ——— ①
 iii) $\frac{\pi}{4} = r\theta$ ——— ①

$\pi = \frac{1}{2} r^2 \theta$ ——— ②

sub ① into ②

$$\pi = \frac{1}{2} r \frac{\pi}{4}$$

$r = 8 \text{ cm}$ $\therefore$ sub. into ①

$$\theta = \pi/32$$

$$b) 2 \sin 2\theta - 1 = 0$$

$$2 \sin 2\theta = 1$$

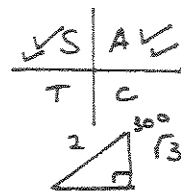
$$\sin 2\theta = \frac{1}{2}$$

acute 2θ is 30°

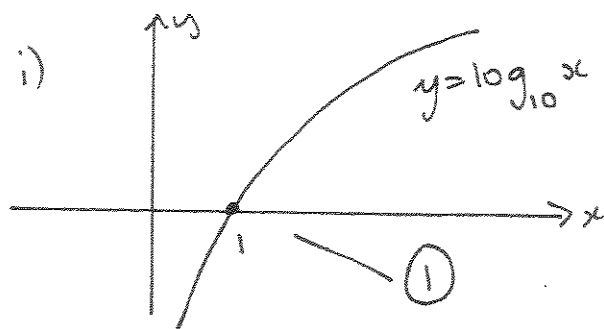
$$\therefore 2\theta = 30^\circ, 150^\circ, 390^\circ, 510^\circ$$

$$\theta = 15^\circ, 75^\circ, 195^\circ, 255^\circ$$

④



c) i)



ii)

x	1	1.5	2	2.5	3
y	0	.176	.301	.398	.477
	F	y ₁	y ₂	y ₃	L

①

$$\text{iii) } \int_1^3 \log x \, dx$$

$$= \frac{.5}{3} [0 + .477 + 4(.176 + .398) + 2 \times .301]$$

$$= .56 \text{ (2 dec places)}$$

QUESTION 6

$$a) i) \alpha + \beta = \frac{3}{2} \text{ ——— ①}$$

$$ii) \alpha\beta = -\frac{4}{2} = -2 \text{ ——— ①}$$

$$\begin{aligned} \text{iii) } (\alpha+1)(\beta+1) &= \alpha\beta + (\alpha+\beta) + 1 \\ &= -2 + \frac{3}{2} + 1 \\ &= \frac{1}{2} \text{ ——— ①} \end{aligned}$$

$$\begin{aligned} \text{iv) } \alpha^2 + \beta^2 &= (\alpha+\beta)^2 - 2\alpha\beta \\ &= \left(\frac{3}{2}\right)^2 + 4 \\ &= 6\frac{1}{4} \text{ ——— ①} \end{aligned}$$

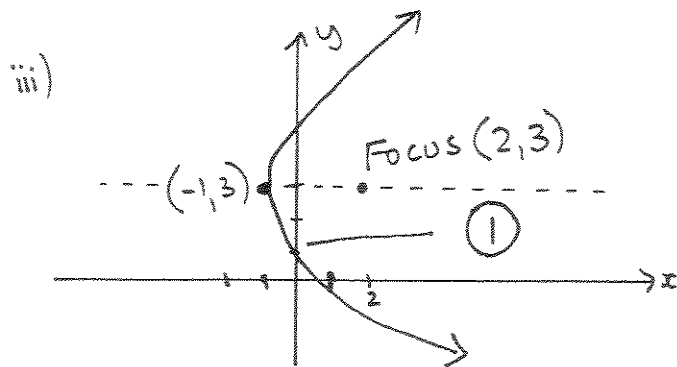
b) $y^2 - 6y - 3 = 12x$

i) $y^2 - 6y + 9 = 12x + 3 + 9$

$(y-3)^2 = 12x + 12$

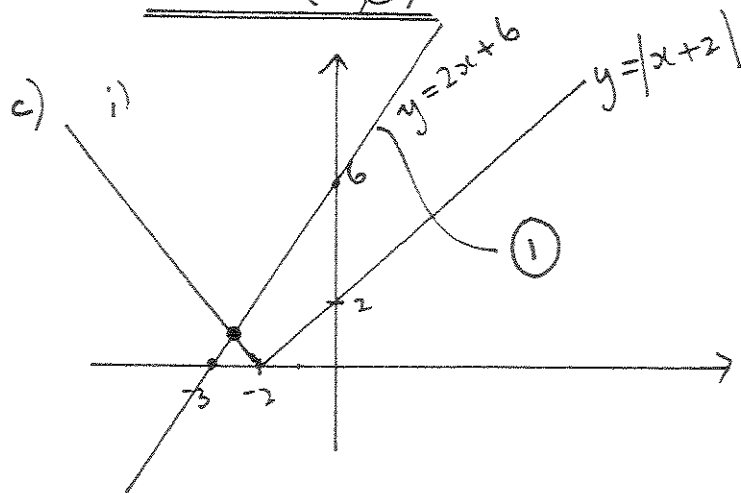
$(y-3)^2 = 12(x+1)$ — ①

ii) Vertex $(-1, 3)$ — ①



$4a = 12 \therefore a = 3$

Focus $(2, 3)$ — ①



ii) Sim eq. $y = 2x + 6$ $y = |x + 2|$

$2x + 6 = x + 2$ or $2x + 6 = -x - 2$

$x = -4$

no a solution
from graph

$3x = -8$

$x = -\frac{8}{3}$

$x = -2\frac{2}{3}$ only
solution

by subst $y = 2x - 2\frac{2}{3} + 6$ — ②
 $y = \frac{2}{3}$

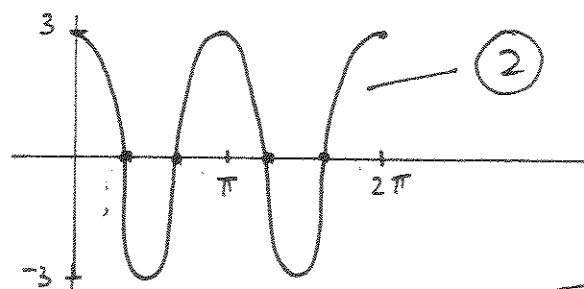
$\therefore$ pt intersection is $(-2\frac{2}{3}, \frac{2}{3})$
(or can be done by subst. into
both equations)

iii) $|x+2| > 2x+6$ — ①
 $x < -2\frac{2}{3}$ from graph

QUESTION 7

a) i) $y = 3\cos 2x$

amp = 3 period = π



ii) 4 points of inflexion — ①

b) $V = \pi \int_0^{\pi/6} \sec^2 x \, dx$ — ①

$= \pi \left[\tan x \right]_0^{\pi/6}$

$= \pi \left[\tan \frac{\pi}{6} - \tan 0 \right]$

$= \frac{\pi}{\sqrt{3}} \text{ unit}^3$

c) i) by Sim. eq. $\sin x = \cos x$

$\therefore \tan x = 1$

$x = \frac{\pi}{4}, \frac{5\pi}{4}$

for A & B
respectively

ii) $\frac{5\pi}{4}$

$A = \int_{\pi/4}^{5\pi/4} (\sin x - \cos x) \, dx$ — ①

$= \left[-\cos x - \sin x \right]_{\pi/4}^{5\pi/4}$ — ①

$= \left[\left(-\cos \frac{5\pi}{4} - \sin \frac{5\pi}{4} \right) - \left(-\cos \frac{\pi}{4} - \sin \frac{\pi}{4} \right) \right]$

$$= \left[\left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \right) - \left(-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} \right) \right]$$

$$= \frac{2}{\sqrt{2}} - \left(-\frac{2}{\sqrt{2}} \right)$$

$$= \frac{4}{\sqrt{2}} \text{ or } \underline{2\sqrt{2} \text{ unit}^2}$$

QUESTION 8

a) $y = x^3 - 3x^2 - 9x + 1 \quad -4 \leq x \leq 5$

i) $\frac{dy}{dx} = 3x^2 - 6x - 9$
 $\frac{d^2y}{dx^2} = 6x - 6$

st. pts $\frac{dy}{dx} = 0 \quad x^2 - 2x - 3 = 0$
 $(x-3)(x+1) = 0$

② $\left\{ \begin{array}{l} \therefore x = 3 \quad (3, -26) \quad y'' > 0 \text{ min} \\ x = -1 \quad (-1, 6) \quad y'' < 0 \text{ max} \end{array} \right.$

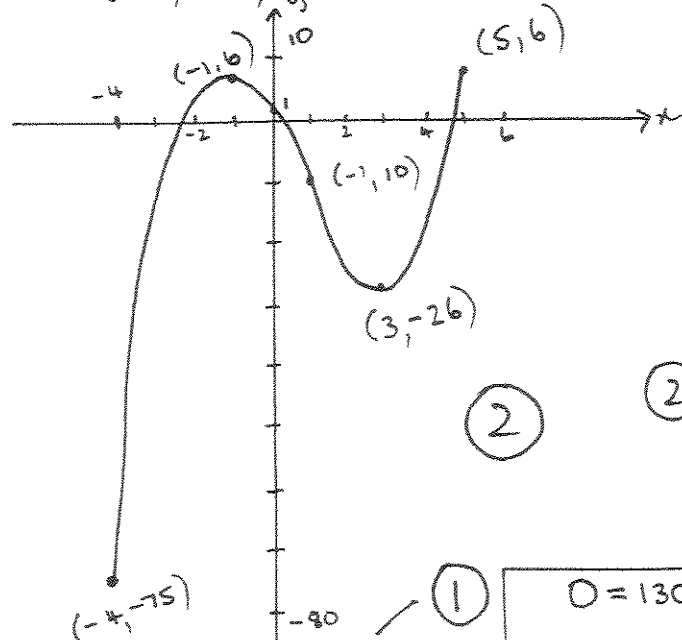
ii) pt inflexion $y'' = 0 \quad x = 1$

① $(1, -10)$ (lies on continuous curve between a max and a min)

iii) end pts

$(-4, -75)$ — ①

$(5, 6)$



iv) min value -75

b) $y = kx^2 - 4\sqrt{3}x + k - 1$

i) $\Delta = (-4\sqrt{3})^2 - 4k(k-1)$
 $= 48 - 4k^2 + 4k$ — ①

ii) +ve def if
 $a > 0$ and $\Delta < 0$ — ①

$\therefore a = k > 0$

and $-4k^2 + 4k + 48 < 0$

$k^2 - k - 12 > 0$

$(k-4)(k+3) > 0$



② $\left\{ \begin{array}{l} \text{ie } k < -3 \text{ and } k > 4 \\ \text{but since } k > 0 \text{ from above} \\ \therefore \underline{k > 4 \text{ only}} \end{array} \right.$

QUESTION 9

a) i) $\frac{d}{dx} (\sin x)^3 = 3 \cos x \sin^2 x$ — ①

ii) $\therefore \int \cos x \sin^2 x dx = \frac{1}{3} \sin^3 x + c$
 do not worry about $+c$

b) i) let A_n be amount owing after n months — ①

$A_1 = 130,000 (1 + \frac{.08125}{12})^1 = \underline{\underline{\$131,056.25}}$
 (before repayment 100 made)

since 9.75% pa $\Rightarrow .8125\% \text{ p.m}$

ii) $A_1 = 130,000 (1.008125)^1 - M$

$A_2 = (130,000 (1.008125)^1 - M) (1.008125)^1 - M$

② $\rightarrow \underline{\underline{= 130,000 (1.008125)^2 - M (1.008125)^1 - M}}$

iii) 13 yrs $n = 156$ loan repaid $A_n = 0$

$A_{156} = 130,000 (1.008125)^{156} - M (1.008125)^{155} - \dots - M$
 $0 = 130,000 (1.008125)^{156} - M [1 + 1.008125^1 + \dots + 1.008125^{155}]$
 $\text{CP } a=1 \quad r=1.008125 \quad n=156$

$$M \left[\frac{1.008125^{156} - 1}{1.008125 - 1} \right] = 130,000 (1.008125)^{156}$$

$$M = \$1473.11 \quad (4)$$

w) Let $M = 1700$ and number of months be n

$$1700 \frac{(1.008125^n - 1)}{0.008125} = 130,000 (1.008125)^n$$

$$1700 (1.008125^n - 1) = 1056.25 (1.008125)^n$$

$$1700 (1.008125)^n - 1700 = 1056.25 (1.008125)^n$$

$$643.75 (1.008125)^n = 1700$$

$$1.008125^n = \frac{1700}{643.75}$$

$$1.008125^n = 2.64077 \dots$$

$$n \log_{10} 1.008125 = \ln(2.64077)$$

$$\underline{n = 120} \quad (2)$$

QUESTION 10

a) i) $6000 + 14000 + 22000 \dots$

AP $a = 6000$ $d = 8000$

$$T_{10} = 6000 + 9 \times 8000$$

$$= \$78,000 \quad (2)$$

ii) $S_n > 1000,000$

$$S_n = \frac{n}{2} (12000 + (n-1) 8000)$$

$$\frac{n}{2} (12000 + 8000n - 8000) > 1000,000$$

$$\frac{n}{2} (12 + 8n - 8) > 1000$$

$$n(4 + 8n) > 2000$$

$$8n^2 + 4n - 2000 > 0 \quad (2)$$

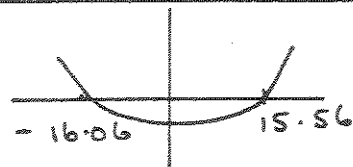
$$2n^2 + n - 500 > 0$$

$$n = \frac{-1 \pm \sqrt{1 - 4 \times 2 \times -500}}{4}$$

$$n = \frac{-1 \pm \sqrt{4001}}{4}$$

$$n = 15.56$$

$$n = -16.06$$



$$\text{since } 2n^2 + n - 500 > 0$$

$$\text{and } n > 0$$

$$\therefore n > 15.56$$

$$n = 16$$

$$\therefore \underline{16 \text{ questions correct}} \quad (2)$$

b) i) $c = 4x - x^2$

$$x^2 - 4x + c = 0$$

$$x = \frac{4 \pm \sqrt{16 - 4c}}{2}$$

$$\therefore x = \frac{4 \pm 2\sqrt{4-c}}{2}$$

$$x = 2 \pm \sqrt{4-c}$$

$$x_1 = 2 - \sqrt{4-c} \quad x_2 = 2 + \sqrt{4-c}$$

$$\therefore (x_2 - x_1) = 2 + \sqrt{4-c} - (2 - \sqrt{4-c})$$

$$(x_2 - x_1) = 2\sqrt{4-c} \quad (2)$$

$$\therefore \underline{A_{\text{rectangle}} = 2c\sqrt{4-c}} \quad (1)$$

ii)

$$u = 2c$$

$$u' = 2$$

$$v = \sqrt{4-c} = (4-c)^{1/2}$$

$$v' = -\frac{1}{2} (4-c)^{-1/2}$$

$$v' = \frac{-1}{2\sqrt{4-c}}$$

$$\frac{dA}{dc} = 2\sqrt{4-c} - \frac{2c}{2\sqrt{4-c}}$$

$$= \frac{2(4-c) - c}{\sqrt{4-c}} \quad (2)$$

$$\frac{dA}{dc} = \frac{8-3c}{\sqrt{4-c}}$$

$$\frac{dA}{dc} = 0 \therefore c = \frac{8}{3}$$

Test max/min

c	1	8/3	3
A'	+		-

max

$$\therefore \text{max } A = 2 \times \frac{8}{3} \sqrt{4 - \frac{8}{3}}$$

$$= \frac{32\sqrt{3}}{3} \quad (1)$$